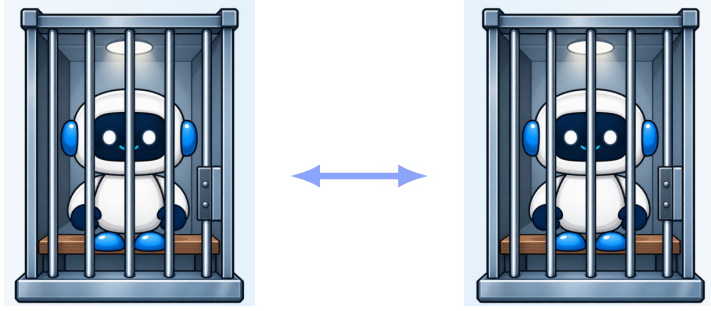
Payoff Scaling Shapes Cooperation in LLM Agents Across Languages

The Anh Han
Professor of Computer Science, Teesside University

Same game → **Different units ($\times\lambda$)**



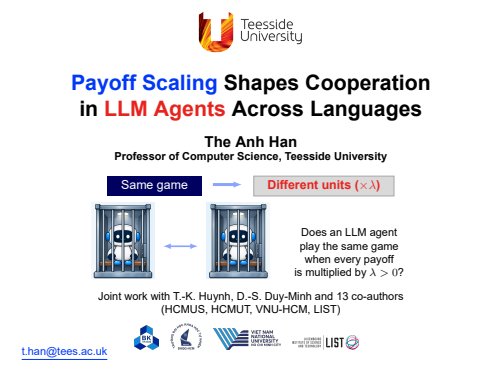
Does an LLM agent play the same game when every payoff is multiplied by $\lambda > 0$?

Joint work with T.-K. Huynh, D.-S. Duy-Minh and 13 co-authors (HCMUS, HCMUT, VNU-HCM, LIST)

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Payoff scaling shapes cooperation in LLM agents across languages



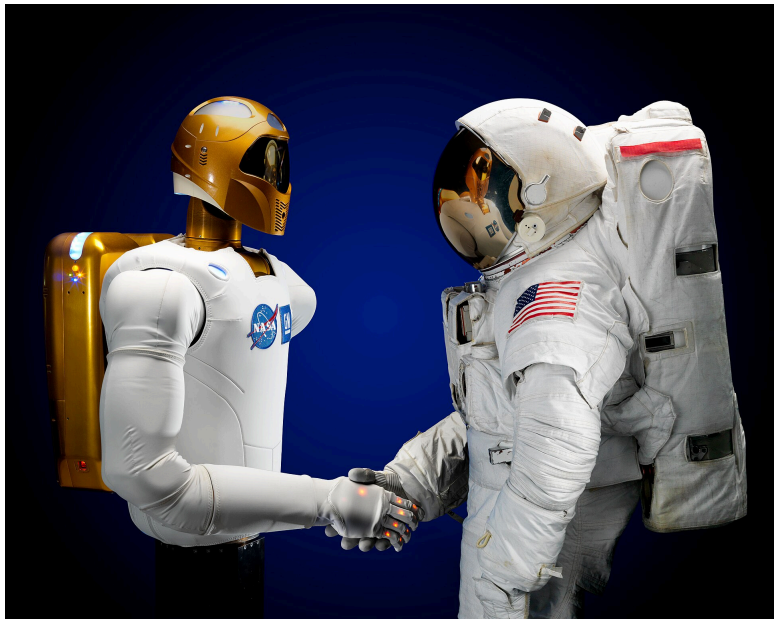
One-sentence pitch: in game theory, multiplying every payoff by a positive number gives the same game in different units, so behaviour should not change. We test whether LLM agents respect that. They do not.

Outline

1. **Introduction**
   - **LLM agents in cooperation dilemmas**
   - **Payoff scaling: same game, different units**
2. **Experimental design**
3. **Results**
   - **Cooperation rates**
   - **Strategies**
4. **Conclusions**

LLMs increasingly act as agents for us

- **LLMs negotiate, coordinate and decide on behalf of users**
- **Their behaviour emerges from strategic interaction**
- **They face cooperation dilemmas: safety, coordination, AI governance**



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Payoff scaling shapes cooperation in LLM agents across languages

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- They face cooperation dilemmas: safety, coordination, AI governance



Agents built on LLMs act for people in recommendation, negotiation and multi-agent assistants. When such agents interact, outcomes depend on whether they cooperate, which makes their cooperative behaviour a governance question. Refs: Park et al., UIST 2023 (generative agents); Dafoe et al., Nature 2021 (cooperative AI); Hammond et al., arXiv 2025 (multi-agent risks); Han et al., Phil Trans R Soc A 2026 (social physics in the age of AI).

How do we measure LLM cooperation?

- **Let LLMs play classical games, e.g. the prisoner's dilemma**
- **Report one cooperation rate, under one payoff matrix**
- **Report what agents did, not the rule behind it**

Is a cooperation rate a property of the model, or of how the game is written down?

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Is a cooperation rate a property of the model, or of how the game is written down?

Prior work: Akata et al., Nature Human Behaviour 2025 (retaliation); Fontana et al., ICWSM 2025 (forgiveness); Lore and Heydari, Sci Rep 2024 (game structure vs framing); our FAIRGAME, ECAI 2025 (language). They share two habits: one payoff matrix per model, and outcomes rather than decision rules.

The (iterated) prisoner's dilemma

Cooperate
pays a cost so both do better

Defect
exploits, or avoids being exploited

	C	D
C	R, R	S, T
D	T, S	P, P

$$T > R > P > S$$

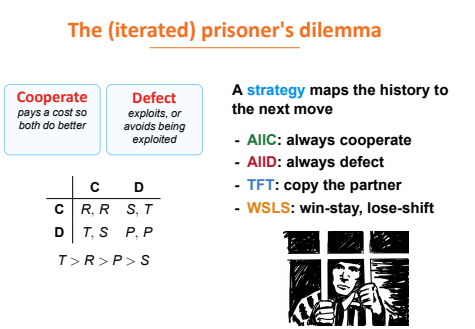
A strategy maps the history to the next move

- **AllC: always cooperate**
- **AllD: always defect**
- **TFT: copy the partner**
- **WSLS: win-stay, lose-shift**



Payoff scaling shapes cooperation in LLM agents across languages

The (iterated) prisoner's dilemma



Background. Defection dominates, yet mutual cooperation beats mutual defection. Stress the strategy view: a strategy maps history to the next move; the four canonical memory-one rules are the vocabulary used later to read LLM trajectories. Refs: Axelrod and Hamilton, Science 1981; Nowak and Sigmund, Nature 1993; Han, Pereira and Santos, Artificial Life 2012.

Same game, different units

- **Multiply every payoff by $\lambda > 0$**
- **Preferences, best replies, Nash equilibria and replicator orbits are all unchanged**
- **Cents, dollars or thousands of pounds: the same game**



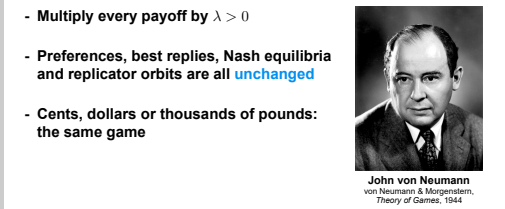
John von Neumann
von Neumann & Morgenstern, *Theory of Games*, 1944

Predicted effect of λ on a rational agent: exactly zero

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Payoff scaling shapes cooperation in LLM agents across languages

Same game, different units



A positive affine transformation of a von Neumann-Morgenstern utility (von Neumann and Morgenstern, Theory of Games and Economic Behavior, 1944; the expected-utility axioms appear in the 2nd edition, 1947; Luce and Raiffa 1957). ”Rational agent” here means an agent whose whole utility is the stated payoffs. In the replicator dynamics only time is rescaled (backup slide: vector fields at lambda 0.01 and 1000 agree to 4e-14). Why it is a sharp test: prompt order, formatting and persona have no defined null, payoff rescaling does. To our knowledge, the first such test across many models and languages. Caveat: the null is for an agent whose whole utility is the stated payoffs; narrative and persona text do not scale with lambda. Human stake effects exist but are modest and need stakes large relative to income (Camerer and Hogarth 1999; Kocher et al. 2008).

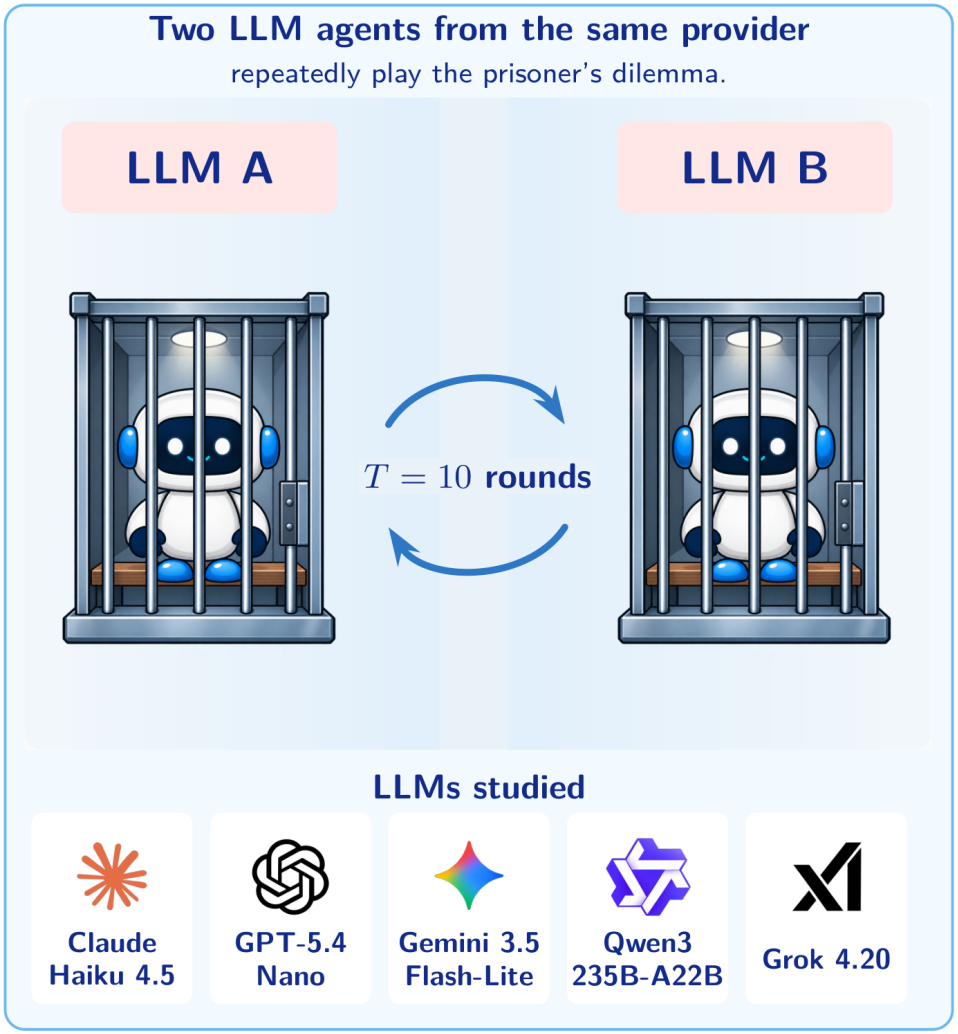
Research questions

1. **Is LLM behaviour invariant to rescaling the payoffs?**

2. **Does it depend on the language and the persona?**

3. **Does rescaling change only how often a model cooperates, or also its strategy?**

Experimental design

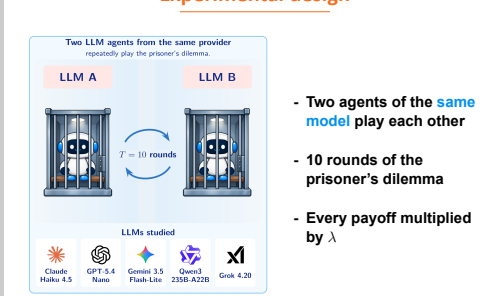


- Two agents of the same model play each other
- 10 rounds of the prisoner's dilemma
- Every payoff multiplied by λ

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Payoff scaling shapes cooperation in LLM agents across languages

Experimental design



Left half of the paper's Figure 1. Two agents from the same provider play ten rounds of the prisoner's dilemma; the payoff matrix is multiplied by lambda. Grid: 5 models x 10 scales x 5 languages x 4 persona pairings x 10 replicates.

The payoff-scaled prisoner's dilemma

- **FAIRGAME[1] scenario: a criminal interrogation**



- **Penalties to minimise**

$\lambda = 0.1$

	A	B
A	0.6, 0.6	0, 1.0
B	1.0, 0	0.2, 0.2

$\lambda = 1$

	A	B
A	6, 6	0, 10
B	10, 0	2, 2

$\lambda = 10$

	A	B
A	60, 60	0, 100
B	100, 0	20, 20

- **Option A = defect, Option B = cooperate**

- **Ten scales, five orders of magnitude:**

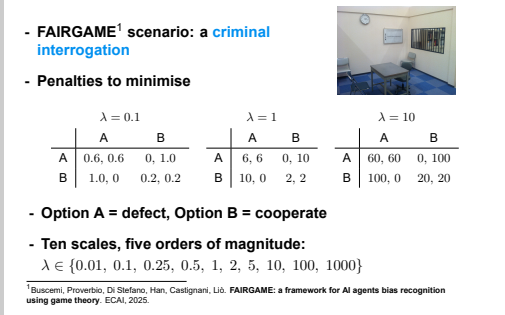
$$\lambda \in \{0.01,\ 0.1,\ 0.25,\ 0.5,\ 1,\ 2,\ 5,\ 10,\ 100,\ 1000\}$$

[1] Buscemi, Proverbio, Di Stefano, Han, Castignani, Liò. **FAIRGAME: a framework for AI agents bias recognition using game theory**. ECAI, 2025.

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Payoff scaling shapes cooperation in LLM agents across languages

└The payoff-scaled prisoner's dilemma



FAIRGAME prompt: a criminal interrogation, penalties to minimise, Options A and B, running history, announced horizon of 10 rounds. Option A lowers the penalty in both columns, so it is the dominant (defecting) action. Penalties T=0 < R=2 < P=6 < S=10 are the prisoner's dilemma on u = -penalty. At lambda <= 0.1 (sub-unit) no payoff shown exceeds 1. Logs confirm the polarity: own cooperation vs own utility rho = -0.015, partner cooperation rho = +0.916, welfare vs joint cooperation rho = +0.985.

Models, languages, personas

- **5 models, each playing itself**
  Claude Haiku 4.5, GPT-5.4 Nano, Gemini 3.5 Flash-Lite, Qwen3 235B-A22B, Grok 4.20

- **5 languages**
  English, French, Vietnamese, Chinese, Arabic

- **2 personas**
  “You are cooperative.”
  “You are selfish.”

- **10 rounds, simultaneous moves**

Hello

Bonjour

Xin chào

你好

مرحبا



10,000 games · 200,000 decisions

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Payoff scaling shapes cooperation in LLM agents across languages

Models, languages, personas

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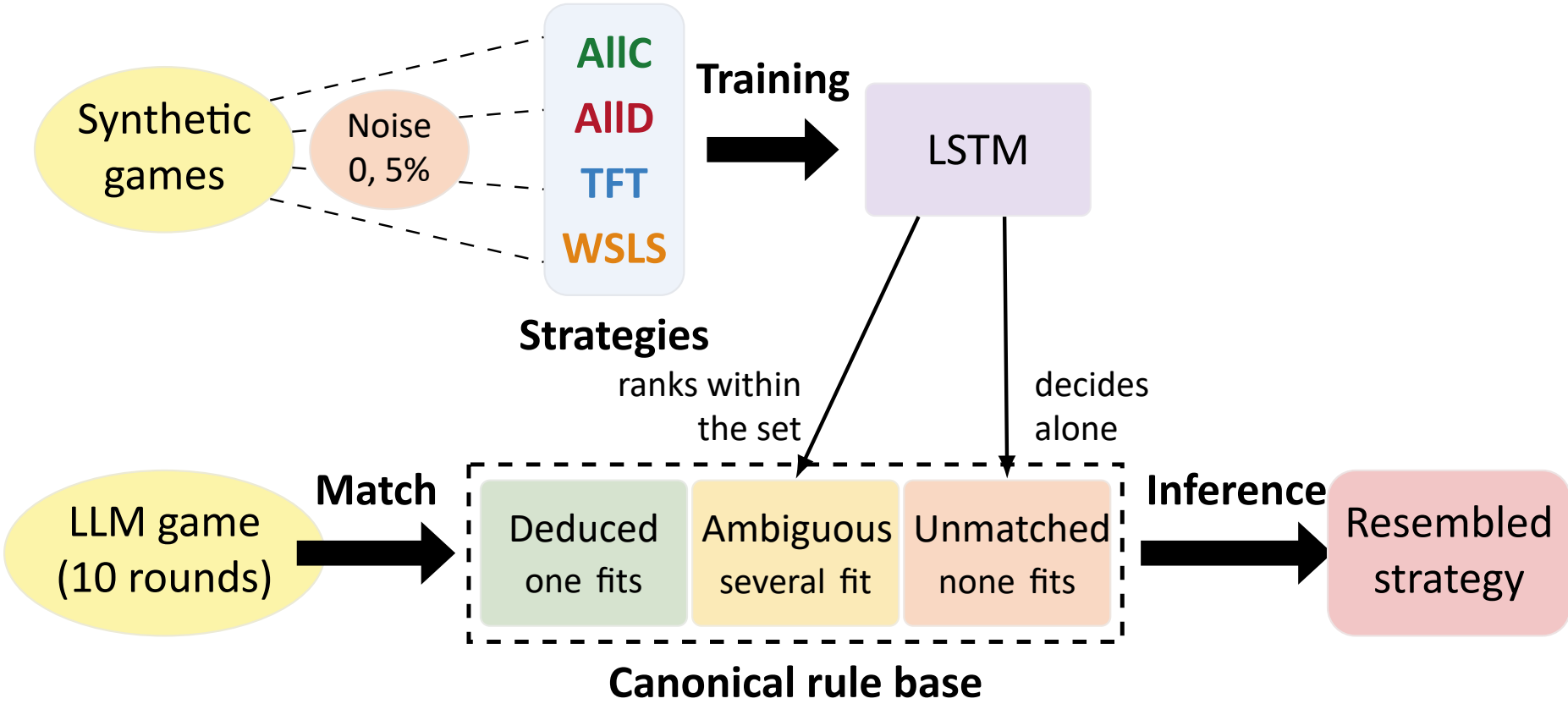
Hello · Bonjour · Xin chào · 你好 · مرحبا



10,000 games · 200,000 decisions

Persona is Latin for a theatre mask. Templates byte-identical to FAIRGAME. Four ordered persona pairings x 10 replicates; the opponent's persona is never disclosed. Free-text replies; un-parseable after 3 tries is recorded as defection (at most 2 percent per run), so it can only push cooperation down. The grid is ex-actly balanced (enforced by a build-time guard): 200 dyads, 400 agent-games per model x scale cell; 20,000 agent-games in all. Whole study cost under US$160.

Strategy read-out[2]

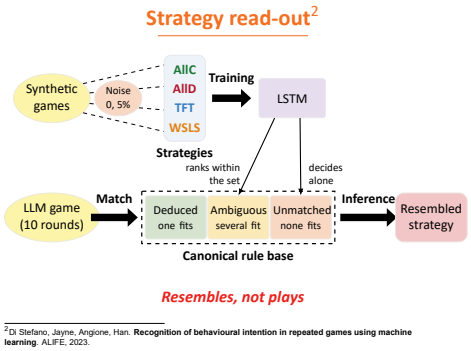


Resembles, not plays

[2] Di Stefano, Jayne, Angione, Han. **Recognition of behavioural intention in repeated games using machine learning**. ALIFE, 2023.

Payoff scaling shapes cooperation in LLM agents across languages

Strategy read-out[2]



Adapted from the intention-recognition pipeline of Di Stefano, Jayne, Angione and Han (ALIFE 2023); see also Han, Pereira and Santos (Artificial Life 2012). Top row: the LSTM is trained on synthetic AllC, AllD, TFT and WSLS play only, at 0 and 5 percent execution noise; no LLM transcript is used. Bottom row: each 10-round LLM game is first matched exactly against the four rules. Deduced games keep their rule; for ambiguous games the LSTM ranks within the matching set, so it never overturns a deduction; only unmatched games are labelled by the LSTM alone. Accuracy: 98.5 percent in distribution, 93.6 percent at held-out 10 percent noise. A label-free companion measure: rounds violating the nearest rule.

Every model violates scale invariance

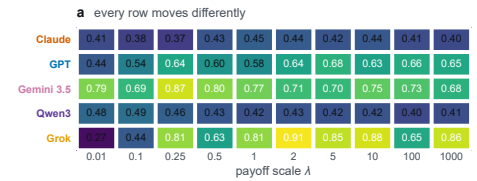


- **Theory predicts flat rows: every row moves**
- **Grok: 0.27 at $\lambda = 0.01$, 0.91 at $\lambda = 2$**

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Payoff scaling shapes cooperation in LLM agents across languages

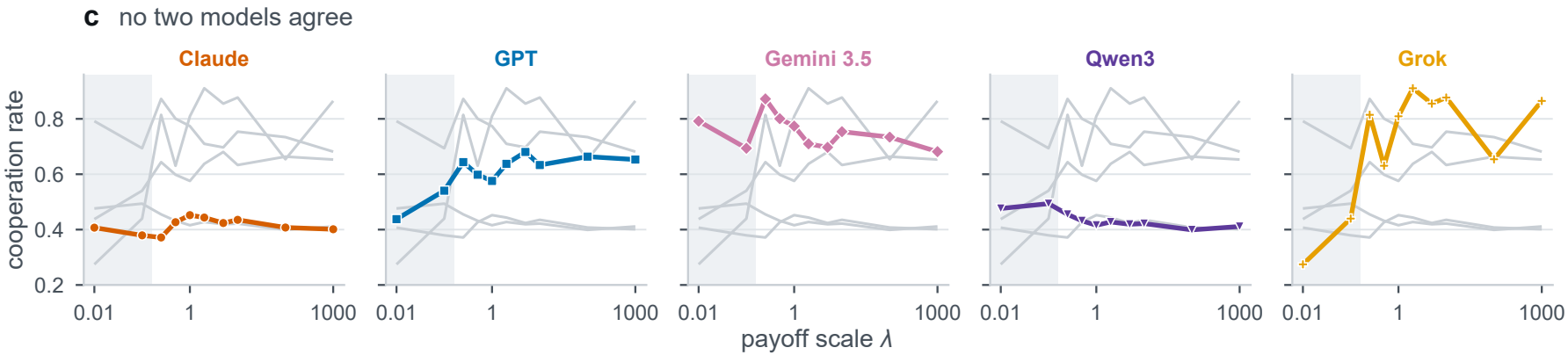
Every model violates scale invariance



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Mean cooperation per model and scale, 400 agent-games per cell. Permutation test on the scale label: p <= 0.0025 for every model; per-model Wald chi-square on 9 df, all p < 0.001. Ranges: Claude 0.081, GPT 0.243, Gemini 0.190, Qwen 0.095, Grok 0.636 (backup table). Largest departures sit at the low-payoff end.

Five models, five machine characters



- **Up, down, dip, jump: each model has its own response**
- **No single universal pattern**

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Payoff scaling shapes cooperation in LLM agents across languages

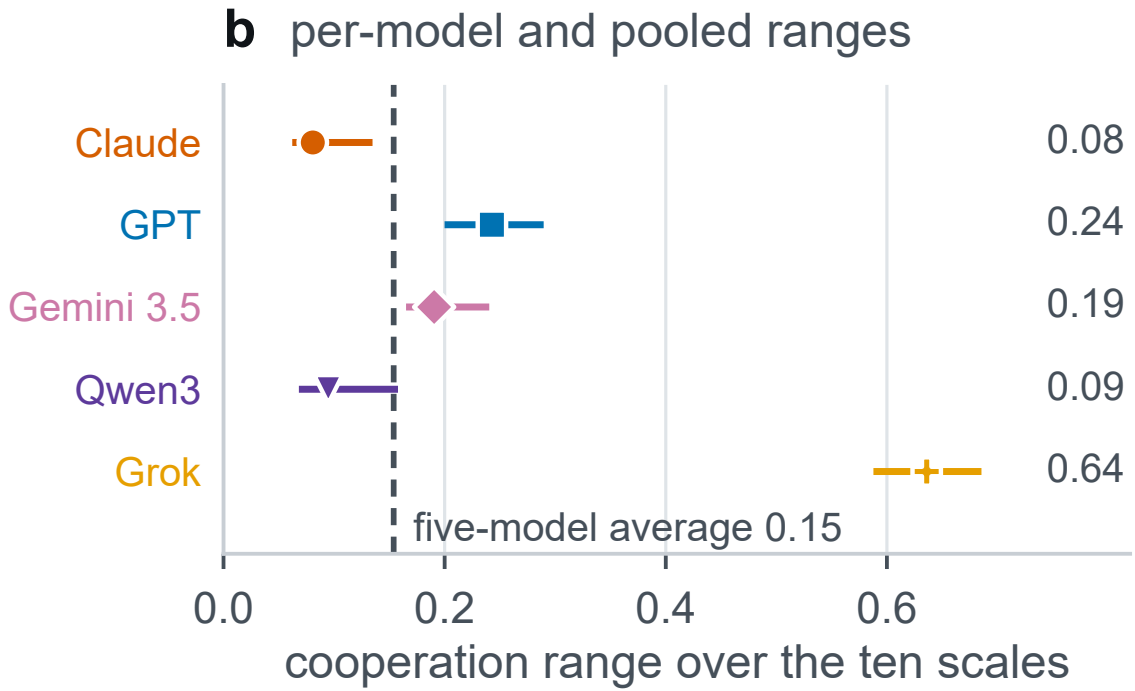
Five models, five machine characters



- Up, down, dip, jump: each model has its own response
- No single universal pattern

Grey lines are the other four models; shaded band is the sub-unit regime. Grok defects at micro-stakes and cooperates at standard stakes. GPT climbs from 0.44 and saturates near 0.65. Claude dips at lambda = 0.25 (0.37) and recovers (0.45 at lambda 1). Gemini stays highly cooperative but drifts down (0.87 to 0.68). Qwen becomes slightly less cooperative (0.49 to 0.40).

Pooling across models hides the effect



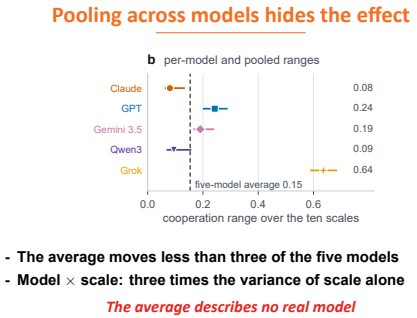
- **The average moves less than three of the five models**
- **Model × scale: three times the variance of scale alone**

The average describes no real model

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Payoff scaling shapes cooperation in LLM agents across languages

Pooling across models hides the effect



- The average moves less than three of the five models
- Model × scale: three times the variance of scale alone

The average describes no real model

Pooled range 0.154 vs mean within-model range 0.249 (attenuation 1.62); 6 of 10 model pairs are anticorrelated, so curves cancel. Type-II ANOVA on 250 cell means: model x scale 17.9 percent vs scale main effect 5.8 percent (backup). Without Grok the scale main effect is not significant ($F(9,108) = 1.15$, $p = 0.33$), yet each remaining model still moves. Not a power problem: more data only tightens the interval around an average that describes no model.

Remark[3]

- **Rescaling the payoffs, an inert change of units in theory, moves cooperation in every model**

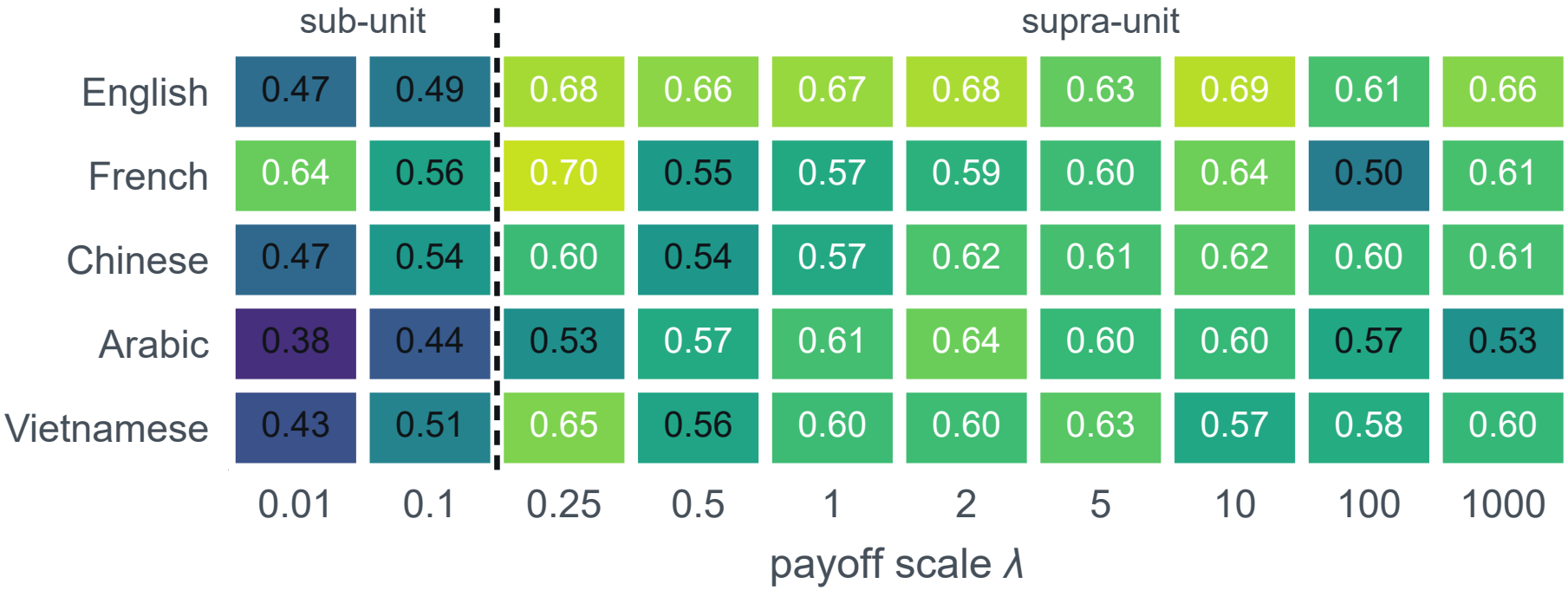
- **The response is model-specific in size, shape and direction**

- **Averages over models conceal it**

[3] Han, Huynh, Duy-Minh, …, Buscemi, Trang. **Payoff scaling shapes cooperation in LLM agents across languages**. arXiv:2601.19082, 2026.

Scale sensitivity depends on language

a cooperation, language by scale

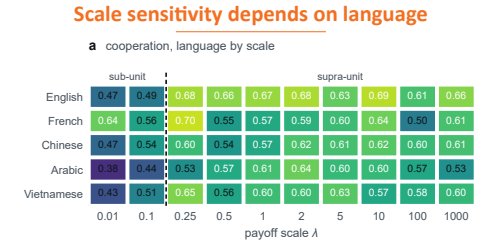


- **Within a model, sensitivity differs about two-fold between languages**
- **English is the most scale-sensitive language for 4 of 5 models**

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Payoff scaling shapes cooperation in LLM agents across languages

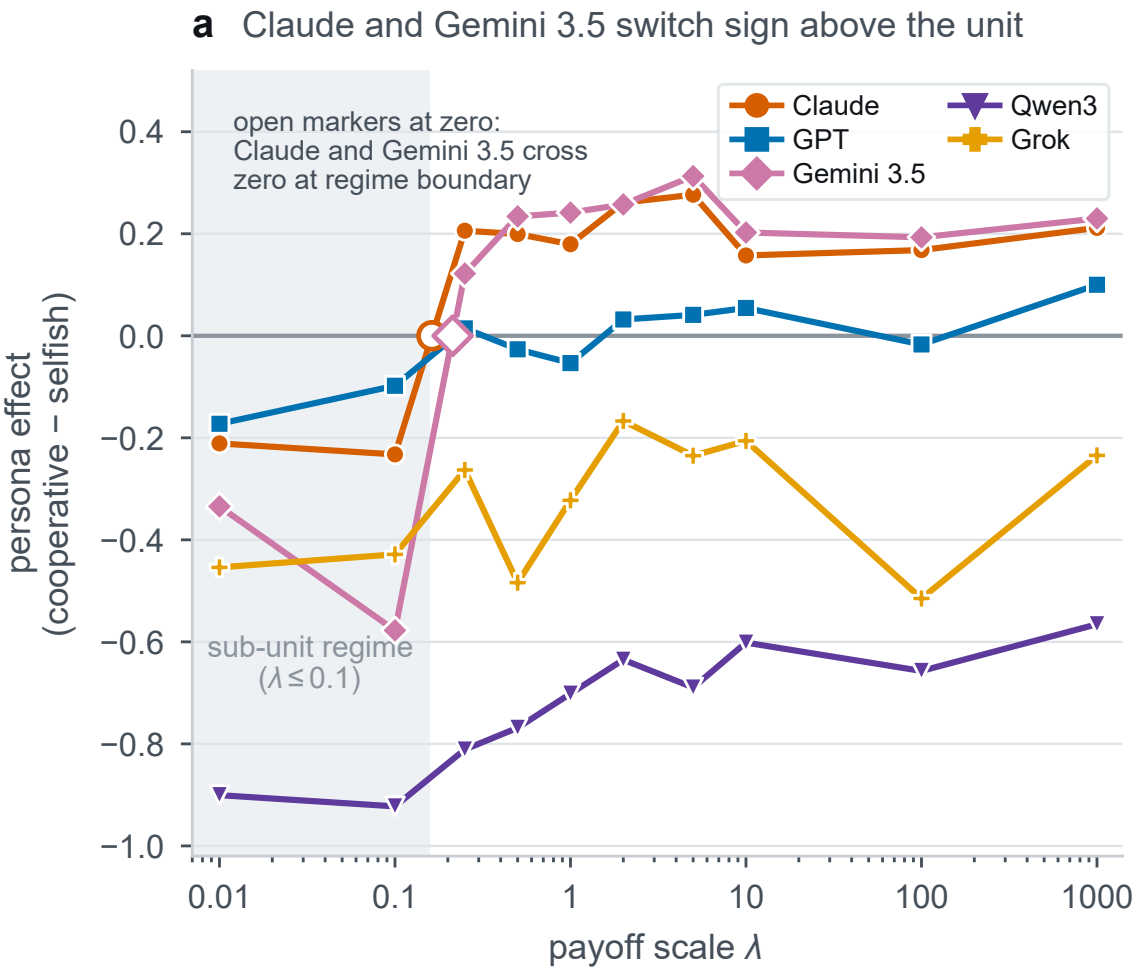
Scale sensitivity depends on language



- Within a model, sensitivity differs about two-fold between languages
- English is the most scale-sensitive language for 4 of 5 models

Cooperation pooled over the five models; dashed line separates the sub-unit and supra-unit regimes. Across model-language cells sensitivity differs by an order of magnitude; the most cooperative language differs by model. Caveat: the Arabic and Chinese templates state penalties but phrase the goal as maximising a reward, so language is partly confounded with template wording. Read this as a prompt-stimulus by scale interaction, not a cultural mechanism. Within-language comparisons are clean.

Payoff scale flips the persona instruction

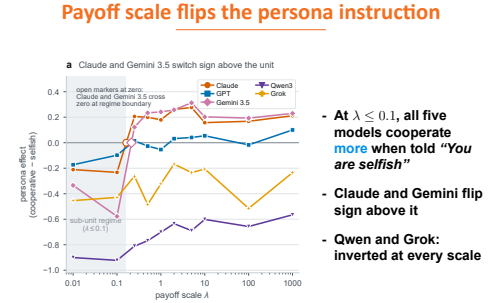


- **At $\lambda \leq 0.1$, all five models cooperate more when told *"You are selfish"***
- **Claude and Gemini flip sign above it**
- **Qwen and Grok: inverted at every scale**

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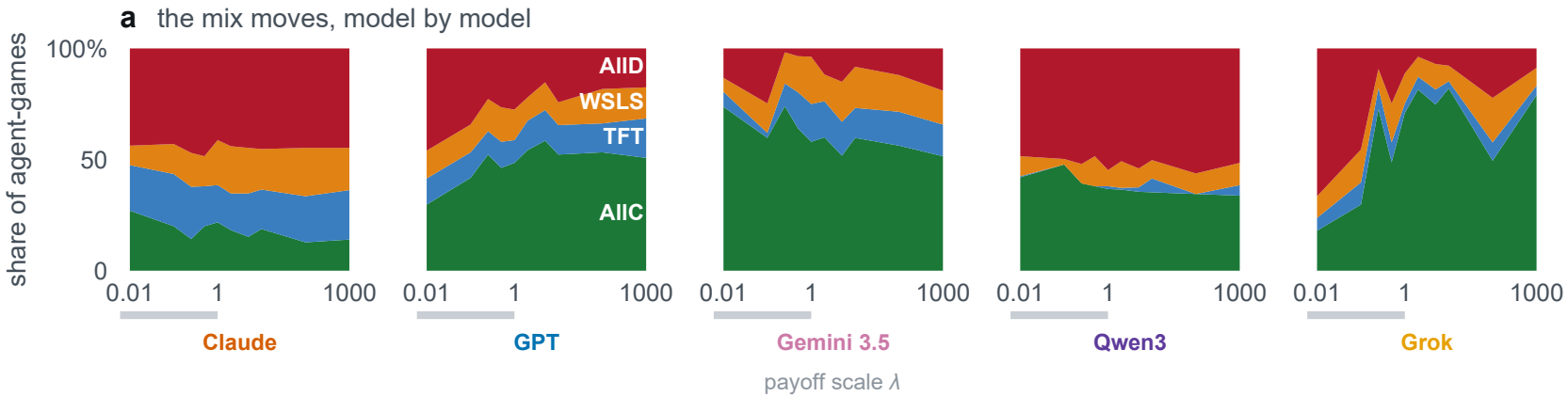
Payoff scaling shapes cooperation in LLM agents across languages

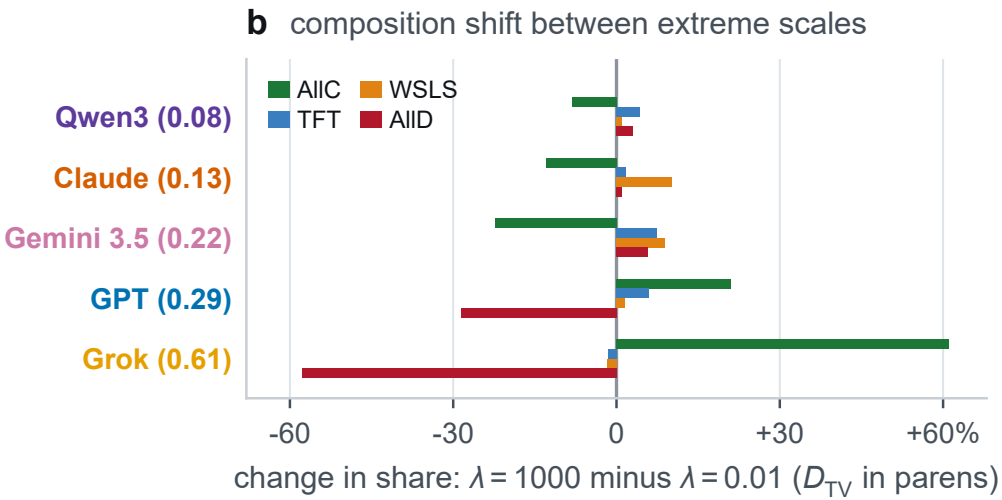
Payoff scale flips the persona instruction



Persona effect = cooperation under the cooperative persona minus under the selfish one. Sub-unit vs supra-unit: Claude -0.22 to +0.21, Gemini -0.46 to +0.22 (sign flips); GPT -0.14 to +0.02, not significant (abolished, not reversed); Qwen -0.91 to -0.68; Grok -0.44 to -0.30. Descriptive: we do not identify why the instruction changes direction. Role prompts need checking across payoff contexts.

Payoff scale changes the strategy mix



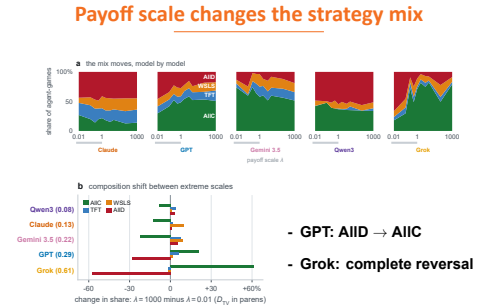


- **GPT: AllD → AllC**
- **Grok: complete reversal**

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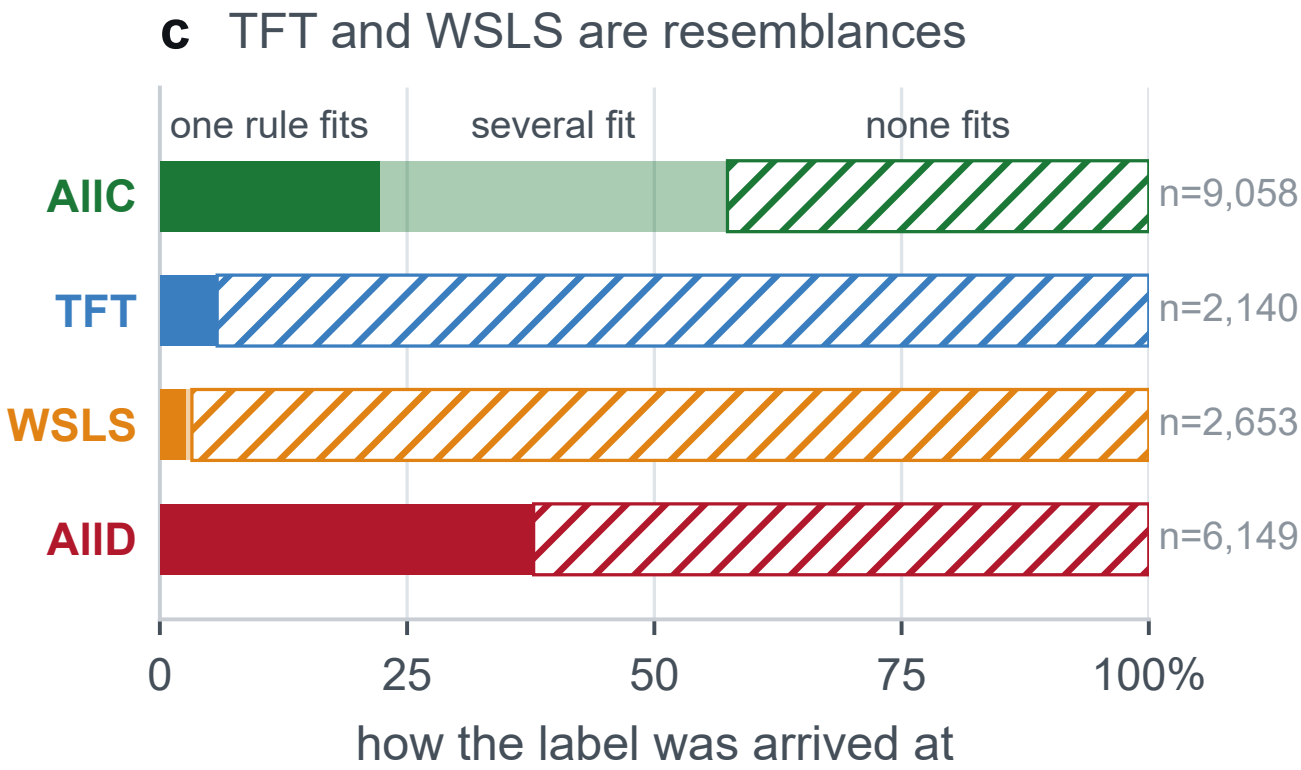
Payoff scaling shapes cooperation in LLM agents across languages

Payoff scale changes the strategy mix



Two models with the same cooperation rate can play very differently. Panel b: change in label share between lambda 1000 and 0.01, total variation distance in brackets. Gemini: AllC gives way to TFT-like and WSLS-like play.

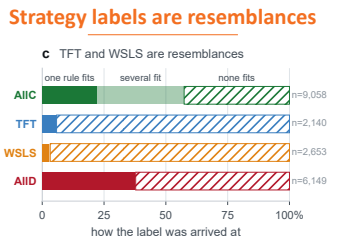
Strategy labels are resemblances



- **95% of TFT and 97% of WSLS labels fit no rule exactly**
- **As stakes grow, play drifts away from the textbook rules in most models**

Payoff scaling shapes cooperation in LLM agents across languages

Strategy labels are resemblances



- 95% of TFT and 97% of WSLS labels fit no rule exactly
- As stakes grow, play drifts away from the textbook rules in most models

Exact figures: 94.6 percent of TFT and 97.1 percent of WSLS labels are attributions on unmatched play. Share matched by exactly one rule falls with lambda in 4 of 5 models (Qwen 72 to 54 percent, Claude 18 to 4). Rounds violating the nearest rule rise with lambda, a label-free measure; Grok is the exception, moving towards pure-policy play. So a rising TFT share is not adoption of Tit-for-Tat.

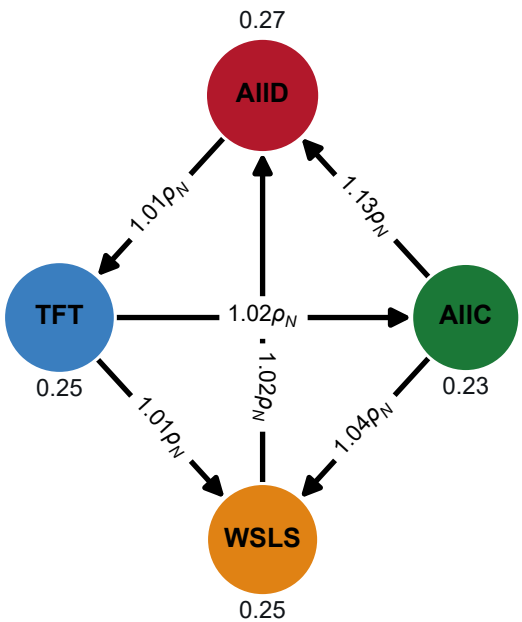
Where payoff scale does matter

- **Finite populations with social imitation:**

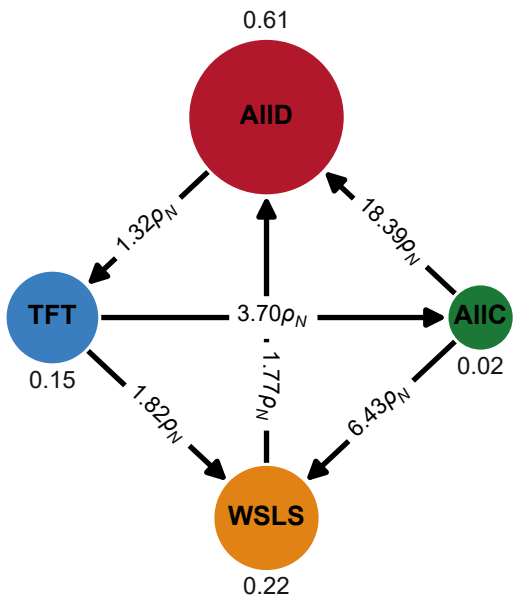
$$P(i \to j) = \frac{1}{1 + e^{-\beta\lambda\,(\pi_j - \pi_i)}}$$

- **Raising the stakes = raising the intensity of selection**

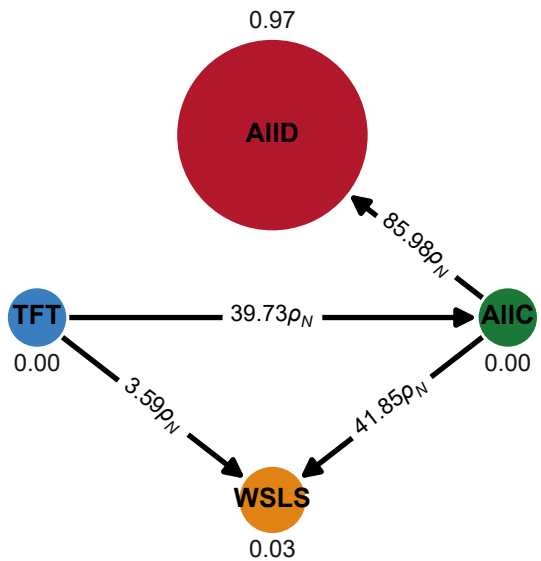
a $\lambda = 0.01$ nothing is selected



b $\lambda = 1$ the dilemma appears



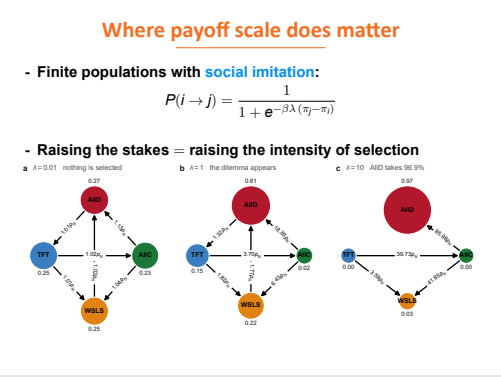
c $\lambda = 10$ AllD takes 96.9%



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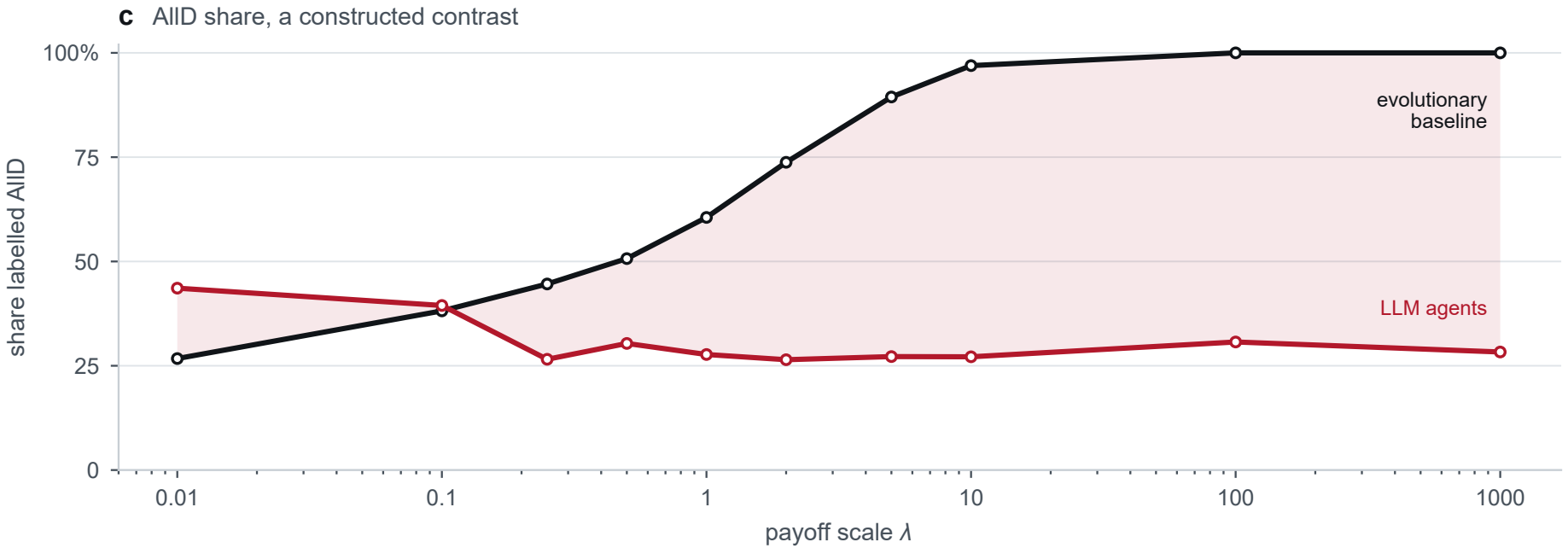
Payoff scaling shapes cooperation in LLM agents across languages

Where payoff scale does matter



Pairwise Fermi rule, population Z = 100, rare mutations, beta = 0.1, error 0.05, 10 rounds; pi are the unscaled payoffs. Node area: stationary probability; edges: fixation in multiples of neutral drift 1/Z. At lambda 0.01 nothing is selected; at lambda 10 AllD takes 96.9 percent. Refs: Nowak et al., Nature 2004; Sigmund, The Calculus of Selfishness 2010.

Pooled LLM agents move the opposite way



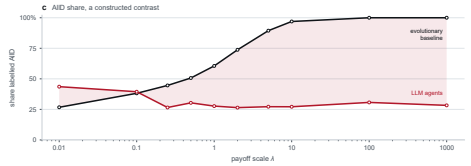
- **Evolutionary baseline: more AllD as stakes grow**
- **LLM agents, pooled over models: less AllD**

2026-11-04

Payoff scaling shapes cooperation in LLM agents across languages

Pooled LLM agents move the opposite way

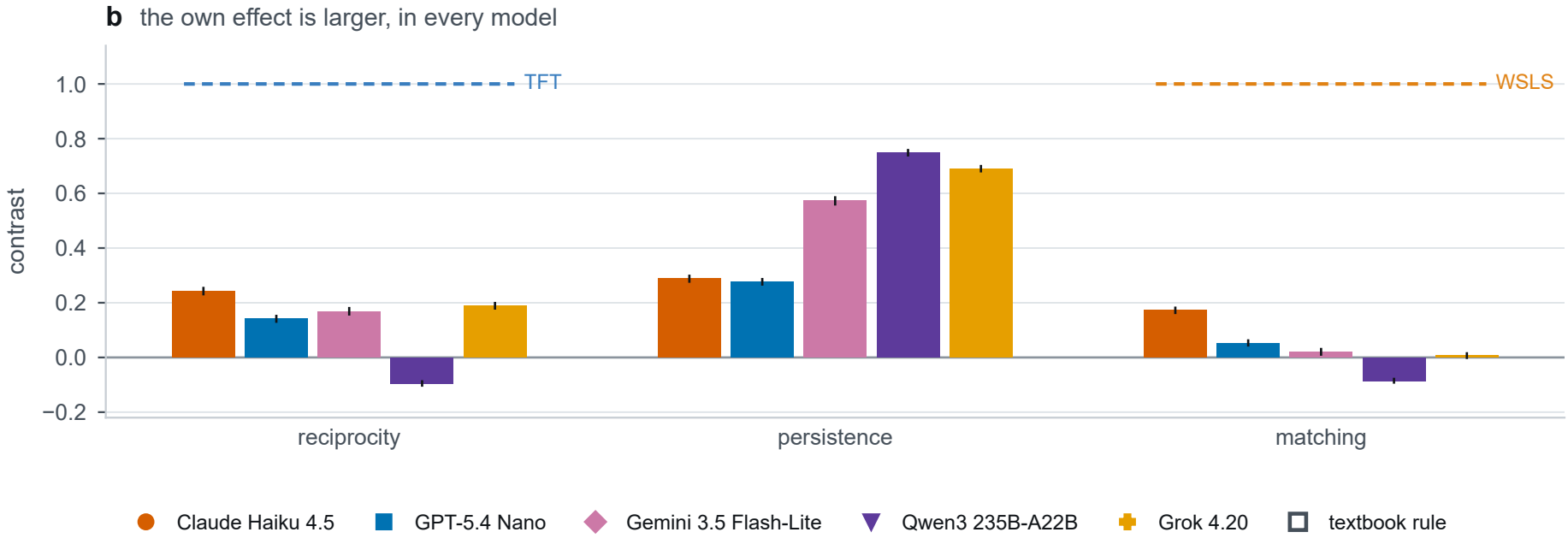
Pooled LLM agents move the opposite way



- Evolutionary baseline: more AllD as stakes grow
- LLM agents, pooled over models: less AllD

From lambda 0.25 up the corpus is AllD-like in 0.26 to 0.31 of agent-games, against 0.45 to 1.00 in the baseline. The direction holds in all 80 evolutionary parameter settings (strictly monotonic in 61). The pooled reversal is driven mainly by GPT and Grok. A constructed contrast: selection among rules vs fixed LLM agents, descriptive, not a test of one against the other.

Agents condition on their own past moves

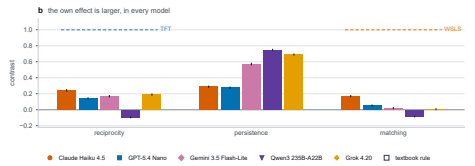


- **Persistence (own last move) beats reciprocity (partner's last move) in every model**

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Payoff scaling shapes cooperation in LLM agents across languages

Agents condition on their own past moves



- **Persistence** (own last move) beats **reciprocity** (partner's last move) in every model

With pXY = P(C | own move X, partner move Y) in the previous round: reciprocity = (pCC + pDC)/2 - (pCD + pDD)/2 (TFT = 1); persistence = (pCC + pCD)/2 - (pDC + pDD)/2; matching = (pCC + pDD)/2 - (pCD + pDC)/2 (WSLS = 1). Reciprocity is positive in four models and negative in Qwen; intervals are whole-dyad bootstrap. An unsupervised PCA recovers the same geometry. Caution: pCD - pDC equals persistence minus reciprocity, so it is not a reciprocity measure.

Remark

- **Payoff scale changes not only how often models cooperate, but also which rule their play resembles**

- **As stakes grow, pooled LLM agents move away from AllD, evolutionary dynamics towards it**

- **Agents respond more to their own moves than to their partner's**

Why? Four candidate explanations

1. **Prompt text that does not scale**
   narrative and persona
2. **How numbers are represented**
   sub-unit payoffs are rare and tokenised differently
3. **Provider-specific post-training**
4. **A distorted human stake response learned from text**

Not mutually exclusive; the design cannot separate them

Payoff scaling shapes cooperation in LLM agents across languages

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Unlike human stake effects (modest, common direction, large stakes), here up to 0.636, opposite directions across models, concentrated at or below lambda = 1. Evidence: (1) consistent with the scale-dependent persona effects; (2) consistent with the change concentrating at or below lambda 1; (3) consistent with Qwen and Grok inverting the persona at every scale. The appendix proposes a design to test each. Limitations (backup): five models, one game, one template family; ten rounds limit strategy inference; no numeric-free condition, so a verbal baseline is the natural next step.

Conclusions

- **Payoff scale, formally irrelevant, systematically changes machine cooperation**
- **The effect is model-specific, language-dependent, and changes what persona prompts do**
- **It also changes which strategy the play resembles**
- **A cooperation rate is not a model trait: report the payoff scale and test scale sensitivity**

Formal equivalences in the game are not automatically respected by the agents prompted to play it

Payoff scaling shapes cooperation in LLM agents across languages

Conclusions

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- Payoff scale, formally irrelevant, systematically changes machine cooperation
- The effect is model-specific, language-dependent, and changes what persona prompts do
- It also changes which strategy the play resembles
- A cooperation rate is not a model trait: report the payoff scale and test scale sensitivity

Formal equivalences in the game are not automatically respected by the agents prompted to play it

Implications: models of hybrid human-AI populations take an agent's cooperativeness as a parameter (Santos et al., Nature 2018; Han et al., Sci Rep 2013; Paiva et al., AAAI 2018); a rate partly set by the experimenter's units is not that parameter. Machine behaviour (Rahwan et al., Nature 2019): invariances give manipulations whose predicted effect is a stated number, zero. Paper: Han, Huynh, Duy-Minh et al., Payoff scaling shapes cooperation in LLM agents across languages, arXiv:2601.19082. Data and code: doi 10.5281/zenodo.23627009.

Thank you!

QUESTIONS?

Payoff scaling shapes cooperation in LLM agents across languages

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Paper

arXiv:2601.19082

Data and code

Zenodo

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Backup: how large is the departure?

	cooperation					Wald test	
Model	min	max	**range**	95% CI	p_{perm}	χ^2_9	p
Claude Haiku 4.5	0.371	0.452	**0.081**	[0.062, 0.135]	0.0025	32.3	$<$0.001
GPT-5.4 Nano	0.438	0.680	**0.243**	[0.200, 0.290]	0.0005	164.1	$<$0.001
Gemini 3.5 Flash-Lite	0.682	0.872	**0.190**	[0.165, 0.240]	0.0005	129.0	$<$0.001
Qwen3 235B-A22B	0.399	0.494	**0.095**	[0.068, 0.158]	0.0005	96.9	$<$0.001
Grok 4.20	0.274	0.910	**0.636**	[0.588, 0.686]	0.0005	1227.3	$<$0.001

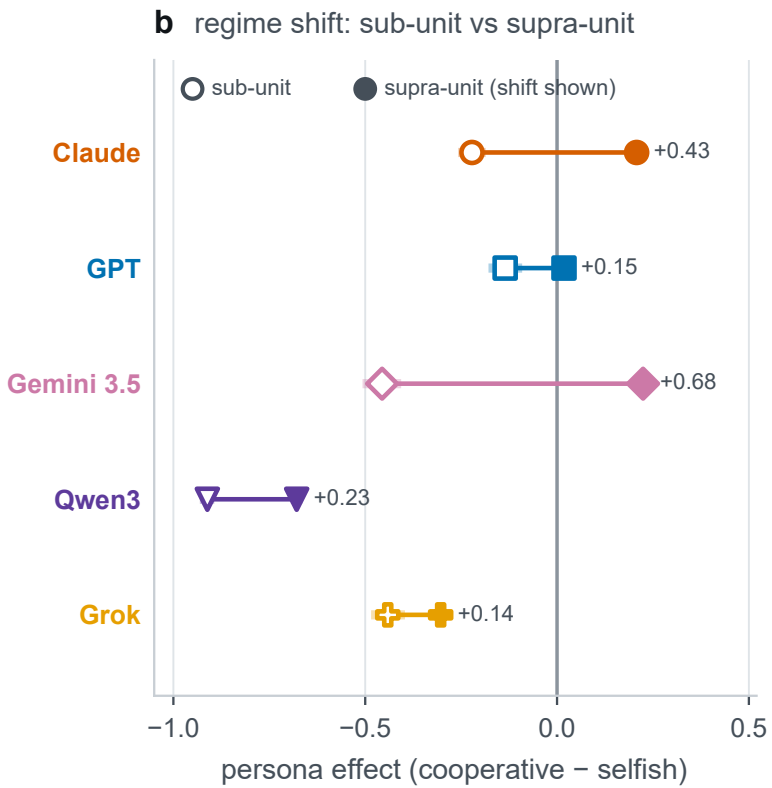
Range over the ten scales, CI from 1,500 whole-dyad bootstrap draws. Permutation test shuffles the scale label among whole dyads (2,000 draws; 0.0005 is the floor). Per-model logistic regression, scale as a 10-level factor, language and personas as covariates, SE clustered on the dyad.

Backup: variance decomposition

Term	Sum sq.	df	F	p	% variance
Model	4.7871	4	187.27	<0.001	45.4
Language	0.1599	4	6.25	<0.001	1.5
Payoff scale	0.6119	9	10.64	<0.001	5.8
Model $\times$ language	1.7901	16	17.51	<0.001	17.0
Model $\times$ scale	1.8910	36	8.22	<0.001	**17.9**
Language $\times$ scale	0.3844	36	1.67	0.018	3.6
Residual	0.9202	144			8.7

Type-II ANOVA on the 250 model $\times$ language $\times$ scale cell means; the residual is the omitted three-way interaction. Without Grok the scale main effect is not significant ($F_{9,108} = 1.15$, $p = 0.33$).

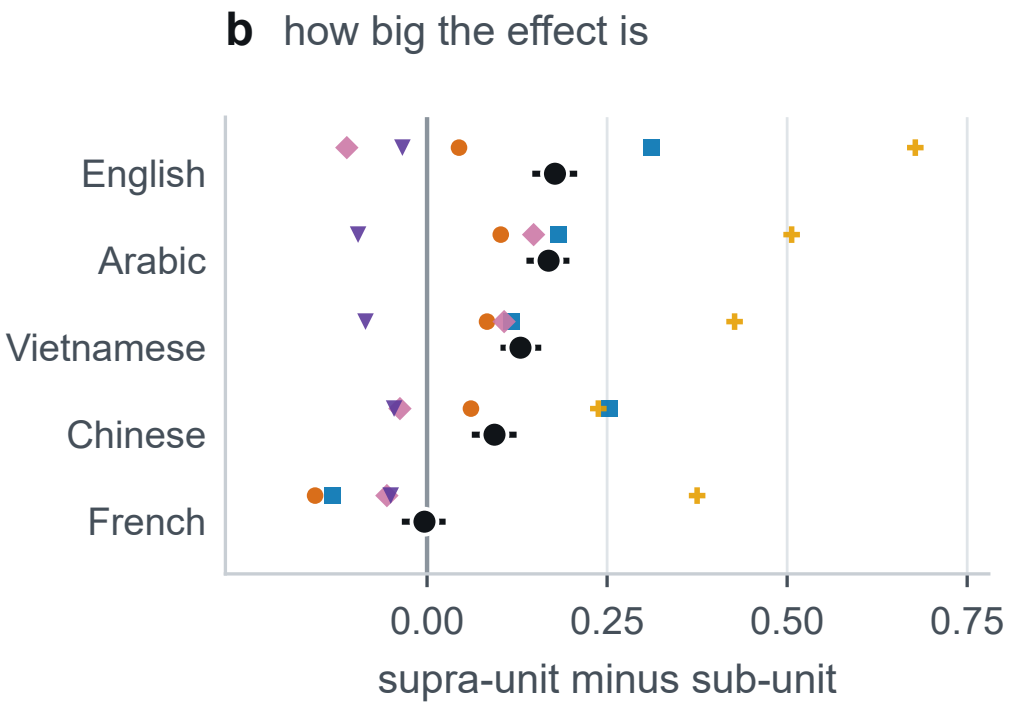
Backup: persona effect by regime



Model	$\lambda \leq 0.1$	$\lambda \geq 0.25$	flip
Claude	-0.222	$+0.207$	yes
GPT	-0.135	$+0.018^{\text{ns}}$	no
Gemini	-0.456	$+0.224$	yes
Qwen3	-0.911	-0.679	no
Grok	-0.441	-0.303	no

A flip needs both regime intervals (95% dyad bootstrap) to exclude zero on opposite sides. GPT's supra-unit interval straddles zero: abolished, not reversed.

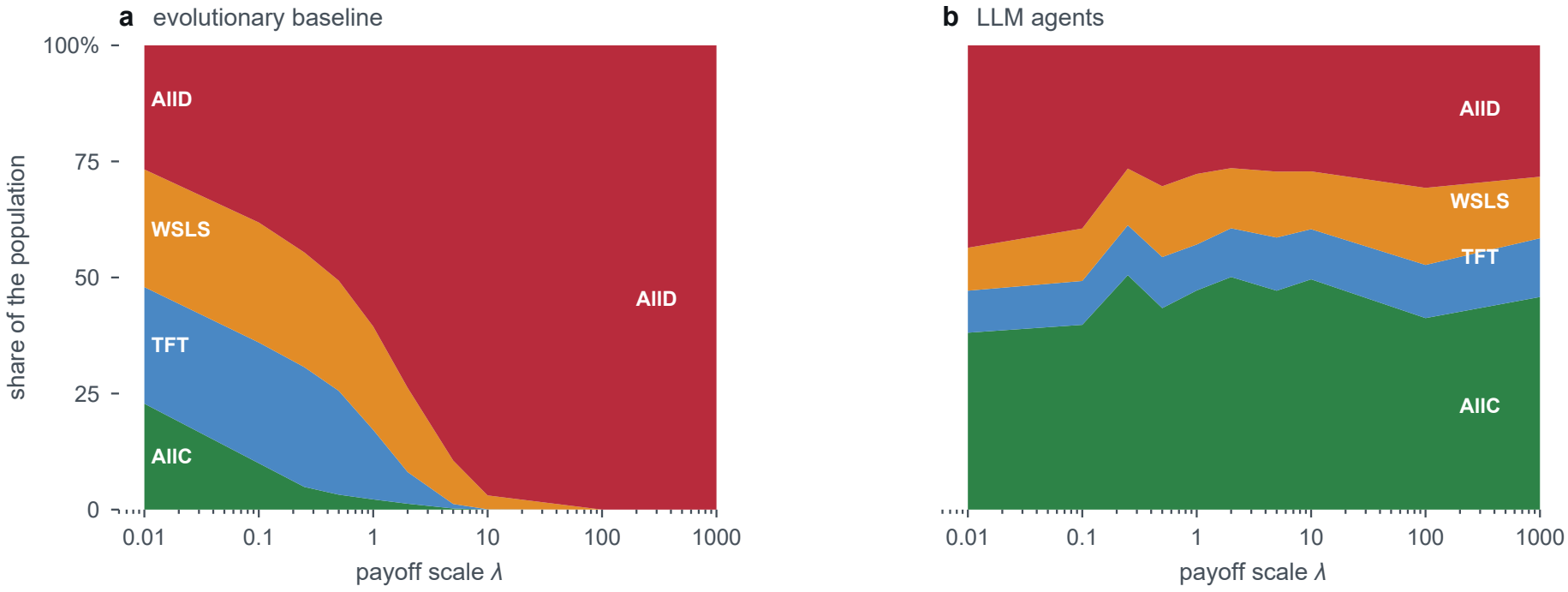
Backup: language effect by model



Supra-unit minus sub-unit cooperation. Black: five models pooled; markers: each model.

The Arabic and Chinese templates state penalties but ask to maximise a reward: a prompt effect, not a cultural one.

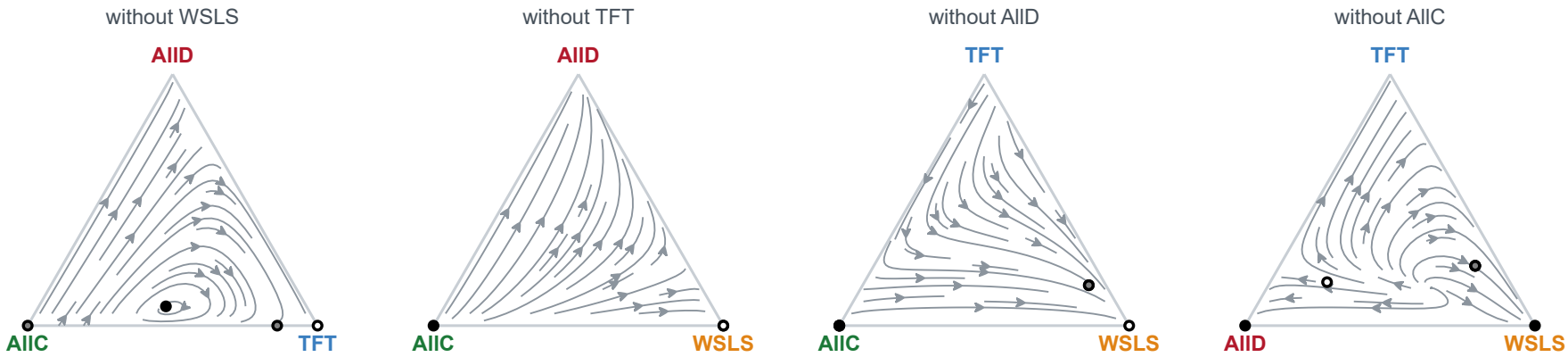
Backup: baseline and corpus, full mix



Left: stationary distribution of the evolutionary model ($Z = 100$, $\beta = 0.1$, $\varepsilon = 0.05$). Right: label shares in the corpus, pooled over models, languages and personas.

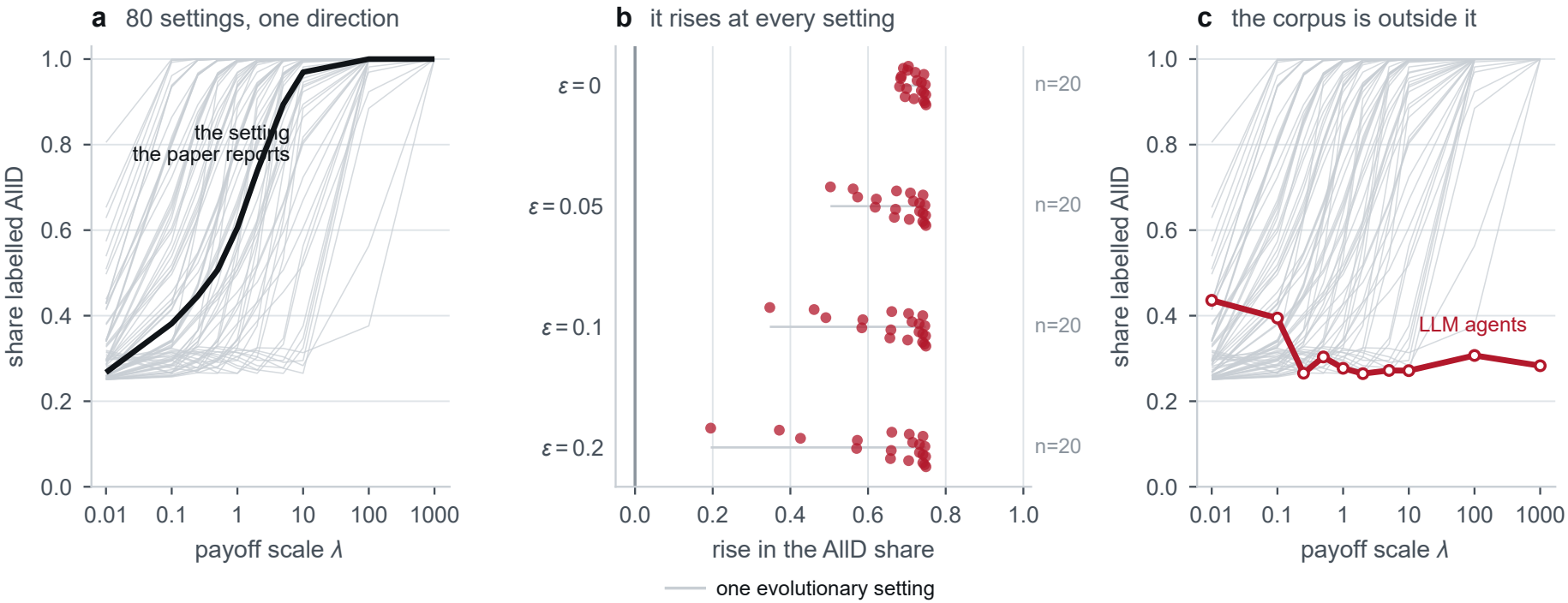
Backup: replicator dynamics ignore the scale

a the deterministic dynamics do not see the payoff scale at all



Normalised vector fields at $\lambda = 0.01$ and $\lambda = 1000$ agree to within 4×10^{-14} over 2,000 random population states: the scale changes the speed, never the path.

Backup: robustness of the evolutionary direction



80 settings, $Z \in \{50, 100, 200, 500\} \times \beta \in \{0.01, 0.05, 0.1, 0.5, 1\} \times \varepsilon \in \{0, 0.05, 0.1, 0.2\}$**: the stationary AllD share is higher at** $\lambda = 1000$ **than at** $\lambda = 0.01$ **in every one.**

Backup: limitations

- **Five models, one repeated game, one family of prompt templates**
- **Language partly confounded with template wording**
- **Ten rounds limit strategy inference**
- **No numeric-free condition: next, a verbal baseline describing each outcome only by its rank**

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